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Scalability bottlenecks in entanglement-based satellite quantum key distribution: downlink asymmetry, orbit dependence and finite-key effects

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Space-based entanglement distribution, as the backbone of the quantum Internet, is poised to enable global quantum networking from infrastructure to application platforms. Satellite-based quantum key distribution (SatQKD), with the entanglement-based QKD protocol of Bennett, Brassard, and Mermin (BBM92), promises source-independent security, but faces a trinity of scalability bottlenecks in dual-downlink configurations. Specifically, asymmetric link characteristics are exacerbated by optical ground station (OGS) pair system parameter differences, while orbit-dependent metrics, e.g., transmission window duration, overpass interval, and trajectory repeatability, introduce finite secret key length (SKL) extraction considerations. Further, quantum satellites will be within the range of multiple OGS pairs, which requires efficient scheduling of SatQKD to prioritize links, mitigate cloud cover disturbance, and maximize finite SKL. Here, we develop a co-modeling framework of dual-downlink SatQKD to support orbital parameter selection and performance comparison for asymmetric system-optics (e.g., aperture diameter, extraneous counts) parameters. We clarify the distinct performance of the SKL extraction methods for Sun-synchronous orbits (SSO) and polar orbits. We further incorporate the finite BBM92 SKL analysis into downlink scheduling optimisation to truncate adverse data collection periods and explore the advantages in flexibility and efficiency for global networking. Our work provides actionable design insights for future entanglement-based global quantum communications.

Spaceborne quantum technologies have become a precursor to build a worldwide networked infrastructure of dedicated quantum technologies, i.e., a quantum network, which will enable long-distance secure data transfer[1], distributed quantum sensing[2], and other entanglement-enabled technologies[3,4]. One of the most remarkable space-based applications of quantum mechanics is the global exchange of encryption keys via quantum key distribution (QKD)[5,6]. With significant progress in the level of technology readiness and in-orbit demonstrations of satellite-based QKD (SatQKD)[7–11], SatQKD missions include not only to overcome the direct distance limit of ground-based links[12], but also to connect intercontinental nodes[13], even to be integrated with fiber-based links to join different networks[14,15].

Pioneered by the achievements of the Micius satellite[16–18], satellite quantum communications have energized global interest, spurring several launch missions that are under active development in China[19], Canada[20], Europe[21], Japan[22], Singapore[23] and the United Kingdom[24]. Several proposals[25–35] have been made that aim to be pathfinders for medium-to long-term perspectives that implement different combinations of QKD protocols and quantum channel configurations between space-based infrastructure and ground-based application platforms. Beyond these paradigms, the entanglement-based QKD directly to a pair of optical ground stations (OGS), with the protocol of Bennett, Brassard, and Mermin (BBM92)[36], removes the need to trust the source on the satellite in a dual-downlink SatQKD scenario[37–39]. This paradigm acts as precursors to entanglement-sharing links, which is essential for the development toward a general-purpose quantum Internet[40–43].

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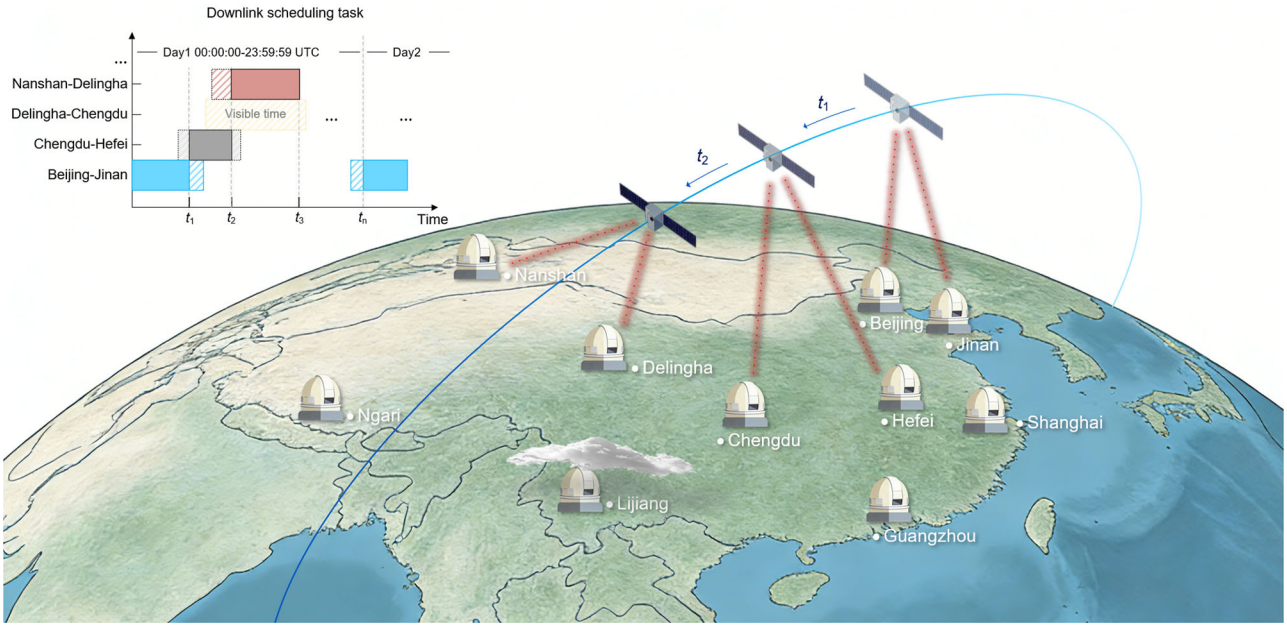


Fig. 1 | Schematic of the dual-downlink SatQKD configuration. The SatQKD configuration consists of an orbiting satellite and multiple OGSs. From t_1 to t_2, the Low Earth orbit (LEO) satellite is at distinct positions during an overpass. It has been apparent that the satellite performing entanglement-based QKD to an OGS pair would encounter shorter transmission windows and higher link loss compared to single downlink Bennett-Brassard 1984 (BB84) configurations. Moreover, asymmetric link characteristics, induced by varied satellite-ground ranges and positions, are exacerbated by system-optics parameter differences between OGSs. In such dual-downlink configurations, the secure key extraction is compounded by processing overpass data, with this complexity arising from variations in channel loss, overpass intervals, and window durations. The Gantt chart illustrates downlink scheduling of SatQKD, reflecting downlink assignment challenges. The overlapping visible time allows downlink scheduling operations, but prioritizing downlinks requires the considerations of finite data block size and changing cloud cover, which is a gap that has not been addressed in networked SatQKD.

Despite these promising prospects, the further implementations of entanglement-based BBM92 for dual-downlink configurations remain particular challenges[44,45] (see Fig. 1). First, different from performing a single downlink QKD to an OGS, establishing and maintaining two downlinks to an OGS pair leads to shorter transmission window durations and higher link losses, further limiting the length of data blocks that could be gathered[46]. More critically, asymmetric link characteristics, induced by varied satellite-ground ranges and positions, are exacerbated by system-optics parameter differences between OGSs and regional cloud cover changes[47,48]. For single downlink SatQKD, a scaling method in which the overpass geometries are represented by different ground track offsets has been proposed to extrapolate performance to other configurations[49]. Accordingly, a generalized dual-downlink SatQKD model is required to support the selection of orbital parameters and comparisons performance for asymmetric system-optics parameters. Second, the selection of finite key extraction methods requires careful treatment of orbit-dependent metrics. Processing overpass data as a single block with or without segmentation presents distinct characteristics[50]. Non-segmentation offers real-time performance but requires sufficient raw data collection, while segmentation yields a sufficiently large block size for asymptotic key extraction, yet introduces higher latency[51]. These characteristics are further compounded as orbital types or altitudes alter channel losses, overpass intervals, and window duration, which determine the selection, but have not been clarified. Third, as LEO satellite constellations for the quantum internet advance, at some times a satellite would be within range of multiple OGS pairs, which creates opportunities for parallel dual-downlink operations[52–55]. This introduces a challenge for SatQKD downlink assignment, particularly in consideration of finite data block size statistics and changing cloud cover[56,57]. Questions such as how to prioritize links based on expected secret key length (SKL) metrics, how to mitigate cloud cover disturbance, and how to maximize total secret key yield within limited window durations remain open, representing a critical gap in scaling entanglement-based SatQKD to networked applications.

We have recently developed a co-modeling framework of downlink Bennett-Brassard 1984 (BB84) QKD channel analysis and orbital propagations towards a role of pre-evaluation[58] and feasibility study[59] in future SatQKD applications. In this work, we extend this framework to entanglement-based BBM92 dual-downlink configurations, addressing the aforementioned gaps. Specifically, our model integrates Sun-synchronous orbit (SSO) and polar orbital propagation, cloud quantification, and entanglement-based QKD channel analysis, which supports orbital parameter selection and performance comparison for asymmetric system-optics parameters. This is similar to the single downlink BB84 SatQKD modeling, using different satellite-OGS ground track distance offsets to characterize satellite overpasses and to extrapolate performance to other configurations. We then compare the annual SKL of two finite-key extraction methods under satellite orbital-dependent metrics, clarifying their applicability for different orbits and providing trade-offs. Building on the scheduling feasibility enabled by the asymmetric downlink loss independence and the merits of single-pass finite-key extraction, we finally formulate the downlink scheduling problem, incorporating finite BBM92 key analysis to prioritize downlinks, mitigate cloud disturbance, and maximize total secret key yield within limited visibility window. Studies of this kind will be valuable to provide a comprehensive methodology and actionable design principles for future entanglement-based SatQKD missions, paving the way towards a robust and scalable global quantum Internet.

This paper is structured as follows. We briefly provide a view of our dual-downlink modeling in section “Modeling framework”, for which the details are presented in section “Methods”. We describe in section “System-optics parameter asymmetry on dual-downlink SatQKD performance” dependence of SatQKD performance on dual-downlink system parameter symmetry and compare the performance of two finite-key extraction methods under different orbits in section “Dependence of SKL extraction on orbit-dependent metrics”. The downlink scheduling with consideration of cloud cover is examined in section ‘Downlink schedule optimisation based on finite key generation performance”. We conclude contributions and

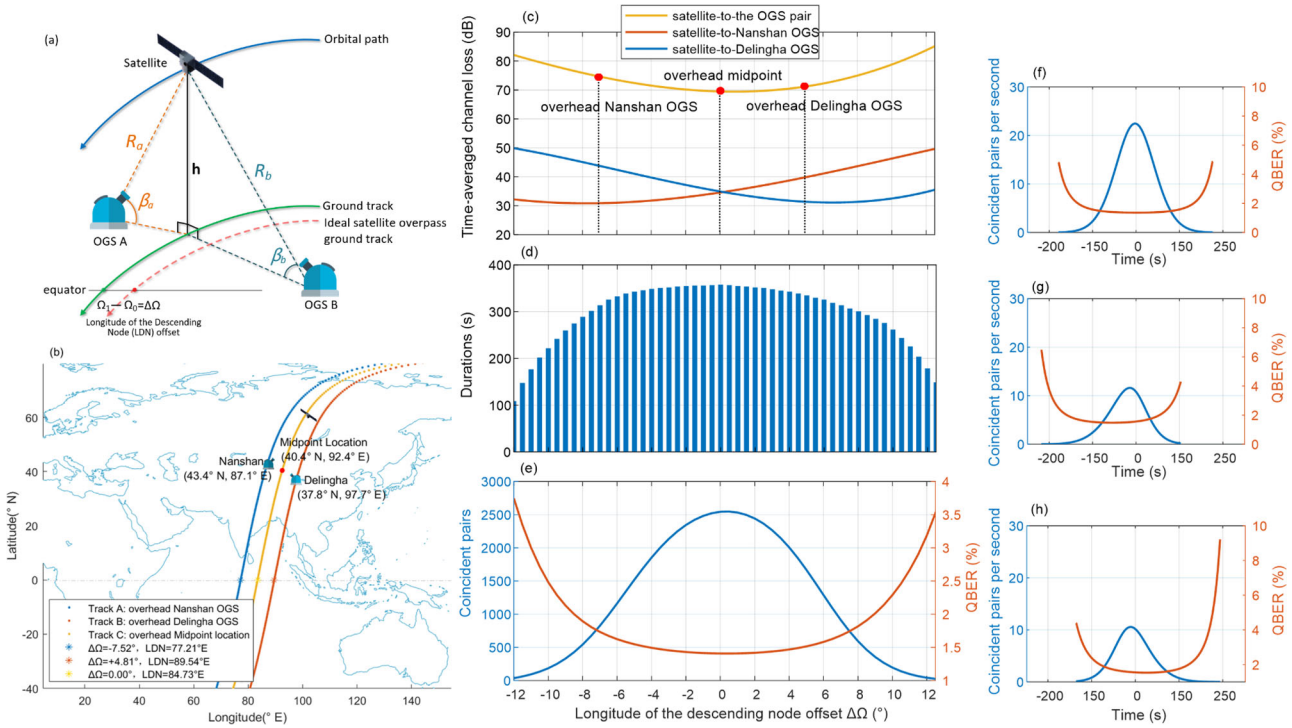


Fig. 2 | Dual downlink entanglement-based SatQKD model. a The general satellite overpass geometry for an orbit of altitude h. In the presence of two OGS locations, the Longitude of the Descending Node (LDN) offsets characterizes different overpasses. The LDN offset is defined as $\Delta\Omega = \Omega_1 - \Omega_0$, where Ω_1 denotes the specific geometric configuration and Ω_0 represents the ideal overpass corresponding to the SSO satellite passing directly overhead the midpoint of the longitudes of the Nanshan and Delingha OGS locations. **b** Ground tracks of the Sun-synchronous orbit (SSO) satellite for different $\Delta\Omega$ corresponding to different overpasses. **c**, **d** Average channel loss and transmission window duration vs $\Delta\Omega$. As expected, the ideal overpass ($\Delta\Omega = 0$) has the longest communication window duration with the lowest average overall two-link loss. The time-averaged channel loss is calculated by first converting instantaneous link loss at each 1-second time bin to linear transmittance, performing time-averaging over the transmission window duration, and finally converting the averaged transmittance back to dB values. For dual-downlink satellite geometry, the two downlink transmittance product is integrated to compute the combined average channel loss. **e** Single overpass data block size (kb) and corresponding QBER vs $\Delta\Omega$. The number of coincident events in **e** was collected at a 1-s interval. **f**–**h** Instantaneous coincident events and QBER for representative LDN offsets, corresponding to the satellite passing directly overhead: (1) the midpoint of the OGS pair, (2) Delingha OGS, and (3) Nanshan OGS. The time $t = 0$ indicates the satellite overpasses the great circle connecting the OGS pair.

outline potential extensions for future global entanglement-based quantum communications in section 'Discussion'.

Results

Modeling framework

Before presenting the detailed results, we first introduce our entanglement-based SatQKD model. What makes the SatQKD applications unique is the dynamic characteristics of optical transmission from satellite to ground, in which the duration of the transmission windows for any pair of line-of-sight OGSs is dependent on the specific orbit of the quantum satellite. To minimize background light's impact on dual downlink SatQKD to an OGS pair at night, we consider a 500 km altitude circular SSO satellite. This orbit ensures that a satellite passes over OGSs at approximately the same local time daily, enabling consistent coverage during critical transits. Dual downlink communication is restricted to the transmission time window, during which both OGSs are simultaneously in view of the satellite, and the quantum link is maintained above a minimum elevation $\beta_{\min} = 10°$. For this, our SatQKD model is integrated with satellite propagations to determine the OGS visibility (see "Methods" section "Integrated with satellite propagations to determine the link geometry parameters" for more details). Elevation and range are calculated as a function of time for different satellite overpass geometries representing different Longitude of the Descending Node (LDN) offsets $\Delta\Omega$ for the satellite overpass (Fig. 2a). As such, the instantaneous dual downlink efficiency $[\eta_a, \eta_b]$ between a satellite and an OGS pair can be given elevation $[\beta_a(t), \beta_b(t)]$ and range $[R_a(t), R_b(t)]$ to quantify the performance of entanglement-based SatQKD during different satellite overpasses.

The instantaneous link loss $\eta_a(\eta_b)$ (dB) characterizes the probability that the individual photons of an entangled photon pair distributed to two parties are detected by the OGS pair, respectively. The total loss generally stems from the optics diffraction, atmospheric absorption and scattering, turbulence-induced fading, and intrinsic system effects. Diffraction loss scales as the inverse-distance squared in the far field, when broaden optical beam signals are received by the finite-aperture telescopes. In practice, acquisition, tracking, and pointing (ATP) system performance and turbulence also determine the received time-averaged spot size. In this work, the satellite transmitter apertures for sending individual photons to the two OGSs are based on parameters reported for the Micius satellite[17], including sub-μrad pointing performance. Given that non-diffraction-limited beam spreading effects, we do not assume diffraction-limited performance, but adopt an identical transmitter beam divergence of 10μrad for both downlink beams. Turbulence-induced profile distortion and beam broadening can be partially mitigated by adaptive optics (AO), while residual turbulence effects are considered due to hardware response speed and real-time processing limits under strong turbulence. For simplicity, we introduce a constant residual turbulence loss into the channel loss model[60]. In addition, the intrinsic system loss is due to single photon detector efficiencies, optical coupling depends on the system design specifications, for which several nominal electro-optical system parameters are listed in Table 1. By altering the diameter of the receiver aperture, the dependence of SatQKD performance on the dual downlink symmetry will be discussed in the next subsection.

The atmospheric absorption and scattering effects are modeled, for which the loss scales exponentially with the straight distance along the

Table 1 | The nominal SatQKD system parameters: the current and expected system parameters relevant to simulation

Definition	Parameter	Value
Orbital altitude	h	500 km
Minimum transmission elevation	$\beta_{\min}$	10°
Receiving aperture diameters	$D_1(D_2)$	80 cm
Transmitter divergence	σ_t	10 μrad
Source wavelength	λ	810 nm
Atmospheric loss at zenith	$\eta_{\text{atm}}^{\text{zen}}$	3 dB
Residual turbulence loss	η_{tur}	0.5 dB
Intrinsic system loss	η_{opt}	4 dB
Pump laser pulse repetition rate	f_s	10^9 Hz
Extraneous count rate	$p_{ec_a}(p_{ec_b})$	5×10^{-7}

In our model, extraneous count probability per coincidence window, including both detector dark counts and background photon counts, is derived from a reported background count rate 500–2000 cps per detector (Moon position dependent) lower bounds by 5×10^{-7}[76], assuming a 1 ns coincidence window. Transmitter beam divergence σ_t, defined as the full-angle at the $1/e^2$ intensity point. An identical value is assumed here for both downlink beams.

elevation β, leading to the path elongation effects at low elevation. This can be estimated by $\eta_{\text{atm}}^{\text{zen}} \csc(\beta)$, where $\eta_{\text{atm}}^{\text{zen}}$ is the atmospheric loss (dB) of satellite-to-ground signals with wavelength λ= 810nm at zenith. Note that $\lambda = 810$ nm is a promising choice for standard fiber-based QKD and supports the future integration of satellite and ground-based fiber QKD systems[14]. However, the greater diffraction of 1550 nm when transmitted through the same sized aperture provides a significant reduction in link efficiency that more than counteracts the slightly better atmospheric transmittance and the turbulence induced beam expansion. Since the dual-downlink geometry further compounds the impact of channel losses, 810 nm is selected for this study. In SatQKD modeling, clear skies over the OGS locations are assumed, setting the same loss $\eta_{\text{atm}}^{\text{zen}}$ at zenith for different OGS locations. Note that we do not explicitly consider transmittance that varies over time by instantaneous spot displacement or cloud disturbance, modeling the average change in link loss due to the variation in range and elevation with time[61]. Consequently, the instantaneous link loss $\eta_a(\eta_b)$ is calculated by discretizing a window duration-long loss change in intervals of one second, thus yielding the average dual downlink channel loss over the window duration. It is remarkable that the cloud coverage induced effect is considered in dual downlink SatQKD scheduling optimization formulation, which enables us to explore the feasibility of performing entanglement-based SatQKD to the OGSs located in those areas with clouds (see "Methods" section "Incorporated with meteorological data to simulate the dual-downlink channels" for details).

Previous studies demonstrate that for single downlink SatQKD with BB84 protocol, the zenith-passing overpass geometry, i.e., the ground track distance offset $\Delta\Omega = 0$, is optimal for secret key length (SKL) generation, since it provides the longest transmission time and the lowest average link loss. However, for the dual downlink SatQKD, the ideal overpass corresponds to the satellite traversing through the great-circle midpoint of the two OGS locations. This can be modeled correspondingly as the Longitude of the Descending Node (LDN) of the SSO $\Delta\Omega$ (Fig. 2b), which is determined from the longitude and latitude of the two OGS locations (see "Methods" section "Ideal overpass geometry"). To this end, the Delingha-Nanshan OGS pair is chosen for the paradigms of the dual downlink entanglement-based SatQKD, where the transmission time window with the average overall two-link loss for the different geometries represented by the LDN offsets $\Delta\Omega$ is shown in Fig. 2c, d, respectively.

Combined with the dual downlink channel model, we further model the entanglement source and detection apparatus of onboard and ground optics to quantify the performance of the entanglement-based SatQKD. In our simulation, we assume that the entanglement is generated via spontaneous parametric down conversion (SPDC) processes. The polarization-entangled photons are split and transmitted to their respective OGS receivers. Due to the significant changes in the loss caused by the satellite overpass (Fig. 2c), the expected coincidence detection rate and QBER have experienced remarkable variations. To model coincidence detection rates, we use link loss $\eta_a(\eta_b)$ combined with the noisy probability of background counts for Delingha OGS and Nanshan OGS to derive the yield of n-photon pairs, which is the conditional probability of coincidence detection given the emission of n-photons (i.e., at least one click on each side). Subsequently, using the pump pulse SPDC source model[62,63], we estimated the overall coincidence probability per pump pulse when using non-photon-number-resolving detectors (see "Methods" section "Finite key analysis of entanglement-based BBM92 SatQKD" for details). In this work, the pump laser pulse repetition rate is 10^9 Hz, with a typical photon pair number per pulse $\mu = 0.053$[64].

Regarding the quantum bit error rate (QBER) calculation, our model incorporates three error contributions: background-induced errors, measurement-imperfection errors, and multi-pair-induced errors. Note that, spectral irradiance could be used to estimate background counts, which is generally a complicated function of atmospheric conditions, the sun or moon sky position relative to the telescope pointing angle, time of day and year, etc. Particularly for polar orbit satellites, the local time of each overpass drifts gradually across the whole year, which leads to continuously changing solar angles and variable background lights. This seasonal variation brings additional complexity to channel noise modeling. In this work, the extraneous count rate $p_{ec_a}(p_{ec_b})$, is assumed constant and independent of elevation, representing the sum of dark and background light counts. Specifically, we take representative values of 5×10^{-7}, which is derived from a reported total extraneous count rate of 100 cps, with a 1 ns coincidence gate[17]. Here, we evaluate the impact of different noise levels via four combinations of extraneous count rates in the subsequent section on system-optics parameter asymmetry, to assess how variations in the total spurious count rate affect system performance. Modeling the dynamic background photon rate with a comprehensive dataset covering different OGS sites, terrains, and weather conditions will be an interesting direction for future research. Figure. 2 e shows the overpass data block sizes and the corresponding QBER during a satellite orbital period for different overpass geometries. The specific receiving processes of the sifted coincident photon-pair statistics are presented (Fig. 2f–h), which allow us to identify data segments detrimental to SKL. In Fig. 2f, when the ground track distance offset $\Delta\Omega = 0$, i.e., the satellite passing directly overhead the midpoint of the OGS pair, the peak coincidence rate occurs since the combined channel loss is minimized. However, the peak coincidence rate can deviate from the midpoint due to geometric effects depending on the overpass geometry and OGS positions. In particular, the peak coincidence rate does not coincide with the minimum QBER. The QBER is determined by the combined effect of distinct, physically motivated error probabilities considered in our model. At the midpoint, high photon detection efficiency maximizes the occurrence probability of multi-pair and measurement errors. As the satellite traverses away from the midpoint, asymmetric channel loss reduces the number of detected source photons, suppressing these dominant error components and leading to a lower overall QBER (see supplementary Fig. 2).

Using the aforementioned model, the satellite orbit parameter setting can be determined, thereby ensuring optimal performance of the entanglement-based SatQKD. Specifically, leveraging the coordinates of an OGS pair, the LDN of the circular SSO satellite can be determined. This enables the satellite to traverse the great-circle midpoint of the two OGS locations, thereby facilitating the maximum data block length with the lowest QBER in a single orbit period. In our simulation, we simplified our model by setting several of the same quantities, such as receiver aperture diameter and background count rate, in the optics setup parameters for the OGS pair. Such differences need also be taken into account, which are particularly important for dual downlink entanglement-based SatQKD operations. In the next section, we highlight the constraints that arise from the system design asymmetry and its implications for the satellite orbit parameter setting.

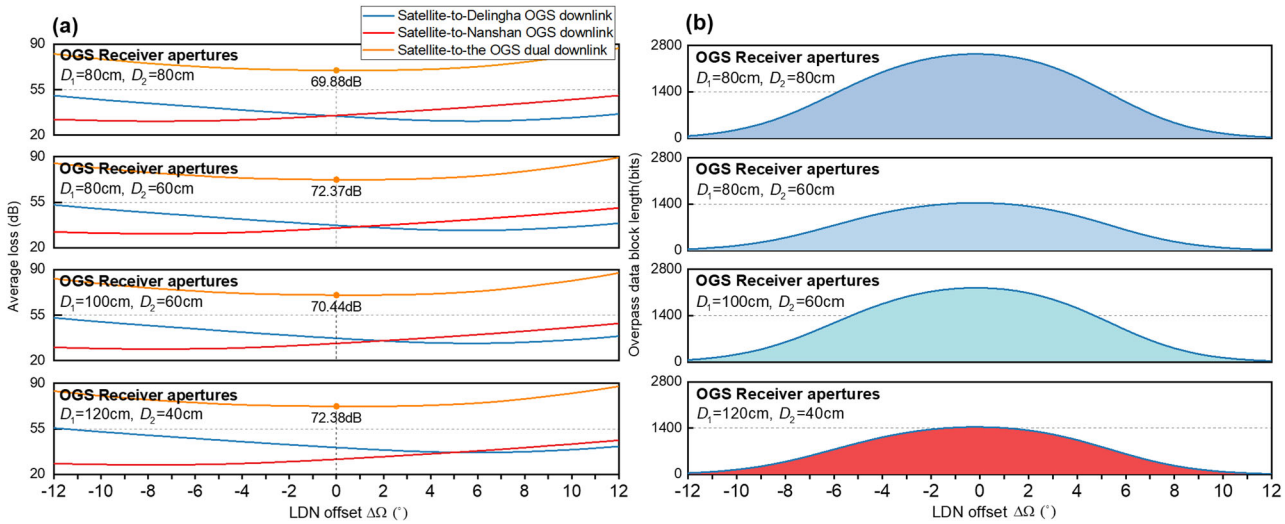


Fig. 3 | Average link loss and overpass data block length vs LDN offset for OGS receiver aperture configurations. Four receiver aperture configurations of diameters (D_1, D_2) of 80 cm/80 cm, 80 cm/60 cm, 100 cm/60 cm, and 120 cm/40 cm for the Nanshan-Delingha OGS pair, respectively. Other optics system parameters are fixed. **a** Average link loss for four configurations. The average link loss for the satellite-to-Delingha OGS, satellite-to-Nanshan OGS, and satellite-to-the OGS pair are represented by the red, blue and yellow curves, respectively. Orange dots mark the two-link loss at the LDN offset $\Delta\Omega = 0$ for each configuration. **b** The corresponding overpass data block length.

System-optics parameter asymmetry on dual-downlink SatQKD performance

The performance of entanglement-based SatQKD depends on dual-downlink change throughout satellite overpasses. As seen in Fig. 2c, d, having a satellite traversing the midpoint zenith of two OGSs is demonstrated to provide the longest transmission time and the lowest average overall two-link loss during an orbital period, thus attaining the largest overpass data block. This is predicated on identical receiver parameter configurations and a maintained dual-downlink symmetry between two OGSs. In contrast, an OGS with a smaller aperture or severe extraneous counts due to light pollution experiences a reduced local signal-to-noise ratio (SNR). The observation prompts the question: can adjusting satellite overpass configuration to enhance local SNR offset the accompanying shorter window durations that can achieve a large overpass data block size?

To address this, four receiver configurations with aperture diameters (D_1, D_2) of 80 cm/80 cm, 80 cm/60 cm, 100 cm/60 cm, and 120 cm/40 cm are selected for the Nanshan-Delingha OGS pair, and other receiver parameters are fixed. For each combination, the average link loss is simulated for overpass geometries defined by constant LDN offsets (Fig. 3a). For asymmetric configurations, For asymmetric configurations, the variant with aperture diameters ($D_1 = 100$ cm, $D_2 = 60$ cm) yields an average dual-downlink loss nearly identical to that of the symmetric baseline ($D_1 = 80$ cm, $D_2 = 80$ cm), a trend consistently observed in the configurations of (D_1, D_2) of 100 cm/60 cm, 120 cm/40 cm. This arises from where the larger aperture in each asymmetric configuration compensates for the higher link loss inherent to the smaller aperture, directly offsetting the additional attenuation suffered by the counterpart with the longer distance and lower elevation.

Furthermore, the corresponding overpass data block lengths are calculated for each aperture configuration and LDN offset (see Fig. 3b). For all receiver aperture configurations with LDN offset $\Delta\Omega = 0$, despite the increased total average link loss due to asymmetric configurations, the longest transmission duration associated with the overpass geometry ensures the largest data block size. Furthermore, when Delingha OGS has a smaller receiver aperture (i.e., $D_2 = 60$ cm), traversing closer to the Delingha OGS zenith (from $\Delta\Omega = 0$ to 6°) leads to a reduction in overpass data block length (Fig. 2b, c). Obtaining different overpass geometries can indeed reduce link loss to the smaller-aperture OGS and align the average dual-link loss with symmetric configurations, yet shorter transmission windows will result in a reduction in data blocks. As such, having a quantum satellite traverse the midpoint zenith of the two OGSs instead of either OGS zenith is optimal for dual-downlink data collection, independent of OGS aperture configurations.

We then consider the dependence of QBER on the SNR. To investigate this, we include four extraneous count rate combinations with (p_{ec_a}, p_{ec_b}) of $10^{-7}/10^{-7}$, $10^{-4}/10^{-7}$, $10^{-5}/10^{-7}$ and $10^{-5}/10^{-5}$ for the Nanshan-Delingha OGS pair. In our simulation, the QBER for each extraneous count rate combination (p_{ec_a}, p_{ec_b}) is estimated for the overpass geometries modeled by the LDN offsets (Fig. 4, in which the symmetric baseline ($D_1 = 80$ cm, $D_2 = 80$ cm) is taken into account to ensure a fair comparison, and different receiver aperture combinations are also introduced to cover more cases. The results show that the difference in extraneous count rates affects the QBER more significantly than the link loss difference. Note that, when the Delingha OGS suffers a higher extraneous count rate (Fig. 4, yellow or pink curve), the satellite traversing away from its zenith further leads to a significant rise in QBER. This is because the effect of extraneous counts is compounded for the higher link loss, thereby diminishing the SNR. Consequently, the ideal overpass geometry, i.e., $\Delta\Omega = 0$, proves suboptimal when the extraneous count rates are asymmetric. These findings indicate the dominant role of downlink SNR asymmetry in determining SatQKD performance and highlight the need to prioritize improvements in light suppression and detector dark count performance alongside OGS aperture specifications.

Dependence of SKL extraction on orbit-dependent metrics

The key generation performance of the SatQKD system depends on the amount of data that could be collected in a finite time. Our results show that the overpass data block size varies with link loss and transmission time window (see Fig. 2). This is particularly evident in entanglement-based SatQKD, for which the transmission time window is demonstrated to be shorter than that of single downlink configurations, and faces additional constraints from asymmetric downlink characteristics. Moreover, satellite orbital variations will induce changes in orbit-dependent metrics, including the average transmission window duration $\overline{T_{\text{window}}}$, the average interval between consecutive overpasses $\overline{T_{\text{interval}}}$, the density of satellite overpasses N, and the average overall two-link loss $\overline{\eta_{\text{total}}}$. These metrics shape SatQKD operations, particularly data collections, and in turn determine the selection of SKL extraction methods.

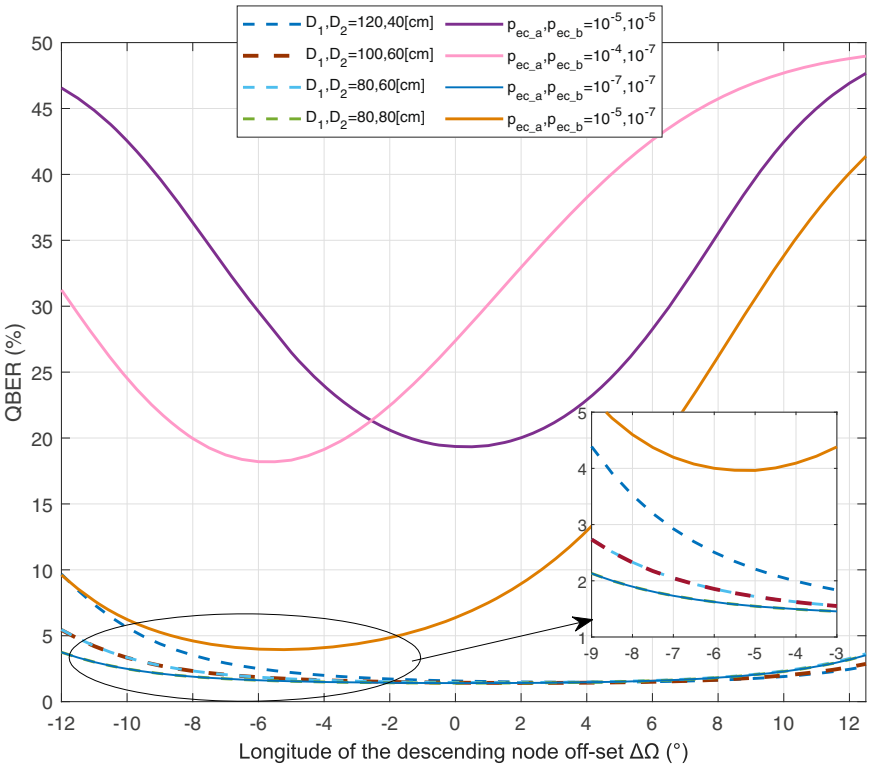


Fig. 4 | QBER vs LDN offset for OGS receiver aperture and background count rate configurations. We include four extraneous count rate combinations with (p_{ec_a}, p_{ec_b}) of $10^{-7}/10^{-7}$, $10^{-7}/10^{-4}$, $10^{-5}/10^{-7}$ and $10^{-5}/10^{-5}$, paired with receiver configurations with aperture diameters (D_1, D_2) of 80 cm/80 cm, 80 cm/60 cm, 100 cm/60 cm, and 120 cm/40 cm. Different curves correspond to QBER variation under the above combined configurations. The inset area highlights the QBER changes for OGS receiver aperture configurations. Small fluctuations indicates that the difference in extraneous count rates impacts the QBER more significantly than that of the link loss difference.

Table 2 | Orbit-dependent metrics for the Delingha-Nanshan OGS pair over a one-year rolling horizon, for Sun-Earth synchronous orbit satellites with an orbital inclination of 97.4065° at different altitudes

h (km)	$\overline{T}_{interval}$	*N*	$\overline{T}_{window}$ (sec)	$\overline{T}_{sum_window}$ (sec)
560.944	42 837	730	363	264 990
888.332	28 317	1095	504	551 880
1257.12	13 969	2190	434	950 460
1676.49	13 727	2190	676	1480 440
2158.56	13 456	2190	946	2071 740

Finite-key security for processing with asymptotic parameters requires partitioning the overpass data into sub-segments of identical statistics for post-processing, for which we refer to this as the *Segmented-block SKL extraction*. Note that it would introduce considerable latency between the start of the first downlink and SKL generation, as sufficient subsequent overpasses are needed. An alternative approach processes overpass data as a single block without segmentation incorporating finite statistics and uncertainties to maintain high levels of composable security (referred to as the "*Overpass-block SKL extraction*"). For small data block lengths, the cost of parameter estimation should be considered, since only a tighter random sampling bound can be utilized for SKL extraction. The "*Segmented-block extraction*" method improves post-processing security at the cost of timeliness, while overpass-block extraction enables low-latency operations. The two methods represent trade-offs between security and timeliness, with distinct characteristics.

To characterize how SKL extraction depends on orbit-dependent metrics, we simulated one-year rolling-horizon metrics for the specialized Sun and Earth synchronous orbit satellites with orbital inclination of 97.4065° at different altitudes for a fair comparison. The one-year simulation span aligns with the Sun synchronism, which, combined with the repeat ground-track properties from its Earth synchronism[65], ensures consistent annual repetition of satellite appearance and transmission patterns for the Delingha-Nanshan OGS pair. Table 2 presents orbit-dependent metrics with different SSO altitudes. We found that higher-altitude SSO satellites promise longer average transmission window durations $\overline{T}_{\text{window}}$, while the density of satellite overpasses *N* remains constant. The reason is that SSO orbital periods align with Earth revolution period therefore relative motion repetition rates are altitude-independent while only transmission window duration varies with altitude. The satellite with an altitude of 500 km is an exception, with a reduced density of satellite overpasses owing to smaller coverage at lower altitude.

We now employ the SKL extraction methods to estimate the long-term average amount of secret key that could be generated with an SSO satellite at different altitudes. We use our entanglement-based SatQKD model to calculate the per-second data for the SSO satellite to the Delingha-Nanshan OGS pair, which is then used to determine positive key generation and compare the performance for different key extraction methods. For the *Overpass-block SKL extraction*, the updated security analysis is considered, which has improved sampling bound to estimate the QBER and phase error performance with small block length. Note that identical SSO satellite overpasses repeat daily, allowing the data from identical passes over several days to be combined to improve SKL. In a one-year period simulation, a fixed number of *n* overpasses per day is specificized by the characteristics of the orbital period, resulting in a total of *N* overpasses. For the *i*th satellite overpass during a day, where $\forall\, i \in n$, $s_i^{\text{overpass_data}}$ and $e_i^{\text{overpass_error}}$ represent the *i*th overpass data (error) block sizes, respectively. Therefore, let $\ell_{\text{SKL}}\left[T\left(s_i^{\text{overpass_data}}, e_i^{\text{overpass_error}}\right)\right]$ denote SKL extracted from a single block, which is formed by aggregating the overpass data blocks over *T* days. The expected annual SKL is given by

$$\ell_{\text{year}} = \frac{365}{T} \times \sum_{i\in n} \ell_{\text{SKL}}\left[T\left(s_i^{\text{overpass_data}}, e_i^{\text{overpass_error}}\right)\right] \quad (1)$$

For the *Segmented-block SKL extraction*, since the overpass data is independent of the loss asymmetry, the instantaneous overall two-link

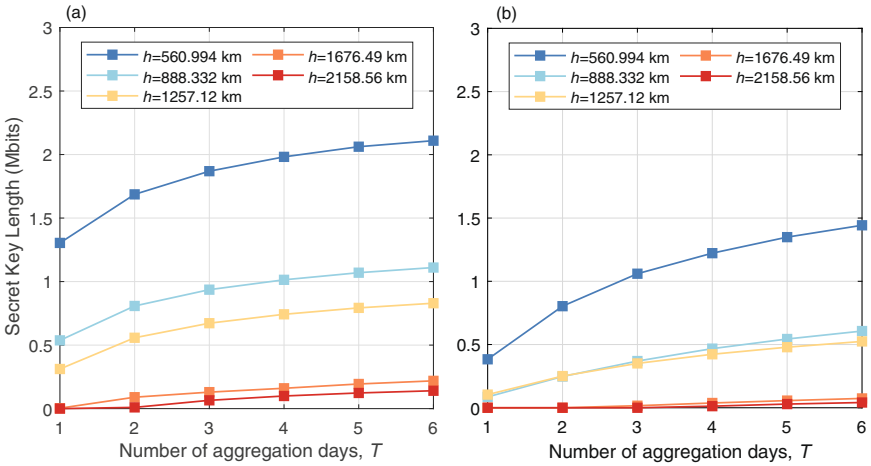


Fig. 5 | Expected annual SKL for the two extraction methods for different number of aggregation days. Five typical orbit altitudes are 560.994 km, 888.332 km, 1257.12 km, 1676.49 km, and 2158.56 km, respectively. Selecting these orbits to achieve an integer sequence of daily Earth-revolution numbers, i.e., 15, 14, 13, 12, 11 for the satellite conforms to the Sun- and Earth-synchronous orbits, while unifying the reference benchmark to simplify data analysis in SatQKD simulations. For a fair comparison, the five orbital altitudes of the polar orbits are the same as those of the SSO. In all cases, we assume that the satellite transmit the polarization-entangled photon pair at a rate of 10^9 pairs/s **a** The expected annual SKL with the *Overpass-block SKL extraction* method. Blocks for post-processing are aggregated over T days, with the number of SSO satellite passes n being identical each day. **b** The expected annual SKL with the *Segmented-block SKL extraction* method. Blocks for post-processing are aggregated from groups of $P = nT$ consecutive overpasses.

losses (per second) from P satellite overpasses are sliced in different segments based on the 1-dB loss-bins. Our adoption of the 1-dB loss-bin aligns with the work in ref. 51, enabling further systematic tuning of photon-pair rates to maximize secure key yield under dynamic channel conditions. As such, the data (error) collected in the corresponding loss bins are segmented into m blocks. Here, we consider the segmented data (error) block $s_j^{\text{segmented_data}}$ ($e_j^{\text{segmented_error}}$) for the segmented block j, where $\forall\, j \in m$ has similar statistics. Let $\ell_{\text{SKL}}\left(s_j^{\text{segmented_data}}, e_j^{\text{segmented_error}}\right)$ denote the SKL extracted from the segmented block j, the expected annual SKL is given by

$$\ell_{\text{year}} = \frac{N}{P} \times \sum_{j \in m} \ell_{\text{SKL}}\left[s_j^{\text{segmented_data}}, e_j^{\text{segmented_error}}\right] \tag{2}$$

Figure 5 shows the expected annual SKL performance of the two methods for SSO altitudes. Unsurprisingly, annual SKL increases for both methods as more days with identical overpasses are included. For the *Overpass-block SKL extraction* in Fig. 5a, we find that at lower altitudes, a significant jump when aggregating data from 1 to 2 days, while the gains from more combined days diminish for all altitudes. This is because single-overpass data may be insufficient for secret key generation, and aggregating data from multiple identical overpasses forms larger post-processing blocks, enabling significant SKL increases as the asymptotic regime is approached. At higher orbital altitudes, blocks aggregated from multiple overpasses remain small, and error counts are more likely to increase under high-loss conditions. Therefore, with no changes to the data volume and the underlying block averaged signal parameters (e.g., QBER), the cost of parameter estimation for post-processing prevents SKL from increasing rapidly. For the *Segmented-block SKL extraction* in Fig. 5b, dynamic variations in overall two-link losses lead to more processing blocks via loss-bin segmentation. Each segmented block is potentially smaller than those aggregated from overpasses with identical geometry, resulting in lower annual SKL than that of the *Overpass-block extraction* method.

The Sun- and Earth-synchronous orbit characteristics enable the *Overpass-block SKL extraction* to generate more keys by aggregating 24 h apart identical overpass data. In practice, the selection of the SKL extraction method should align with satellite orbits and SatQKD operations. As illustrated in Table 2, higher altitudes offer shorter consecutive orbital intervals, making the combination of overpasses more feasible than for low altitudes. Figure 6 shows that *Segmented-block SKL extraction* outperforms *Overpass-block SKL extraction*, especially for polar orbits with higher altitudes, allowing positive key generation. This arises because polar orbits lack daily identical overpasses. The characteristic may restrict the *Overpass-block SKL extraction* method to utilizing data from a single overpass. It can also be expected that downlink variations in SatQKD atmospheric channels further prompt that one prefers to obtain sufficient raw data until a continuous stream of secret key bits could be generated. In practice, securely combining blocks from overpasses of different geometries can still yield a secure key, as is effectively implemented in the *Overpass-block extraction method*, with optimal efficiency for blocks of identical statistics. We also have conducted additional simulations using 0.5-dB, 1-dB, and 2-dB loss bins for comprehensive validation (see Supplementary Fig. 3). The smaller and larger loss bins reveal a trade-off between statistical uncertainty and SKL performance. In particular, 2-dB loss bins outperform 1-dB bins in secret key length. Excessive binning degrades secret key length, which suggests that *Segmented-block extraction method* ultimately could evolve towards *Overpass-block extraction method*. Taken together, conducting comprehensive SKL extraction method investigations in future research, including flexible block-combinations for secure aggregation of blocks with heterogeneous statistical properties could be an essential consideration for quantum communication constellations.

Downlink schedule optimisation based on finite key generation performance

We consider a quantum satellite performing the entanglement-based SatQKD between pairs of different OGS locations (see Fig. 1). For one satellite and large OGS numbers, the potential OGS pairs would only be able to compete for the downlink and communicate sporadically with the satellite. To this end, the downlink schedule can be considered, which is a sequence of tasks of SatQKD to the OGS pairs with discrete time steps (in seconds) that a satellite would execute. Each task in a schedule has a start time when its execution should commence, duration, and the target OGS pair. Notably, the continuous raw data collection process in a transmission duration for an OGS pair could be further segmented due to the overlap of the visible time. Consider that the finite-key security of SatQKD with small block length, secret key extraction should be based on the raw data collected within a single schedule task to avoid tracking and storing data collections. Hence, an efficient downlink schedule needs to be considered based on finite BBM92 SKL analysis.

In this work, we first simulate SatQKD operations of an SSO satellite using Micius ephemeris data for all pairwise combinations among ten OGSs over the week of September 19, 2016. Guided by the analysis in the previous section that demonstrates the high flexibility of SKL overpass extraction, we adopt this key-extraction method for our optimization analysis of the schedule. For an individual satellite overpass, collected raw data are segmented into truncated time blocks with arbitrary start and end instants to

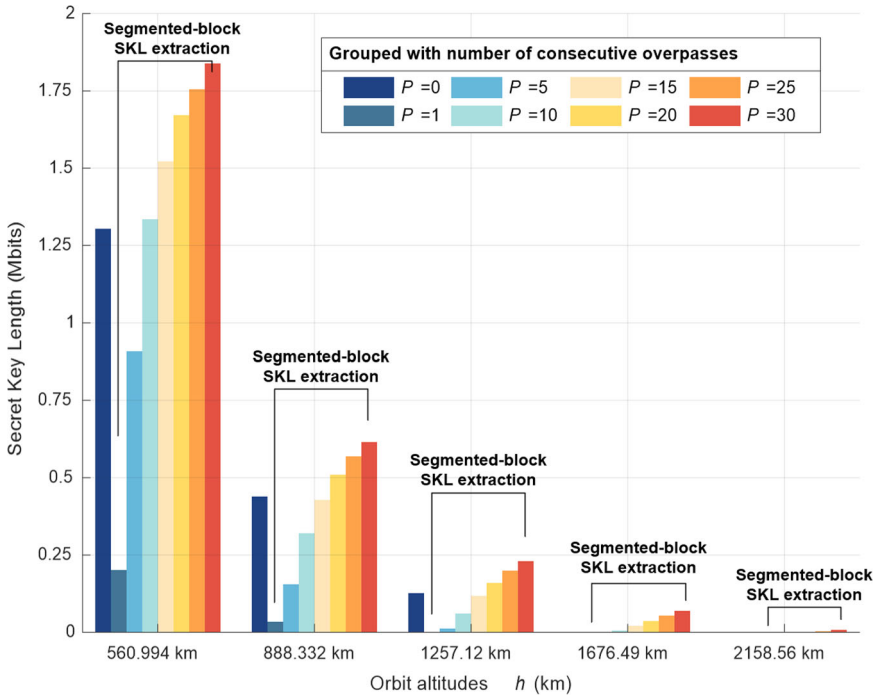


Fig. 6 | Expected annual SKL of the two methods under different polar orbit altitudes. For a fair comparison, five polar orbit altitudes are the same as those of the Sun and Earth synchronous orbits. Blocks for post-processing are aggregated from groups of P consecutive overpasses. Restricted by the non-repeating daily overpass characteristics of polar orbits, in this work, the *Overpass-block SKL extraction* method assumes that only a single overpass data is used for SKL extraction, without aggregating the measurement data from multiple separate overpasses (i.e., $P = 0$). The *Segmented-block SKL extraction*, on the other hand, considers supporting the combination of blocks collected from several consecutive overpass segments to form larger post-processing groups.

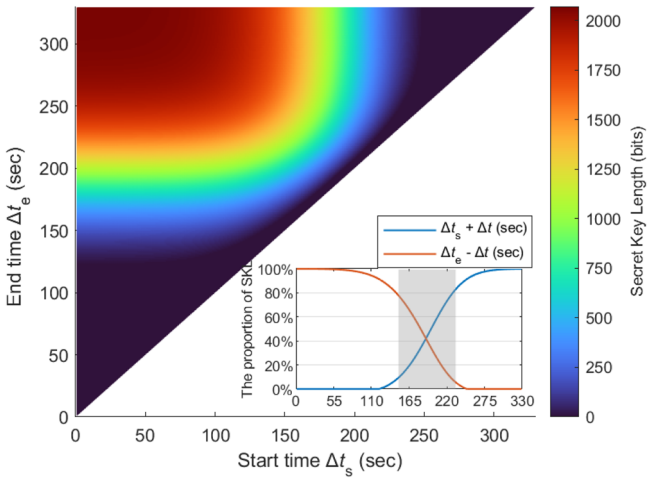


Fig. 7 | SKL vs transmission duration. The finite SKL for the Nanshan-Delingha OGS pair is computed for arbitrary start-to-end time intervals (Δt_s to Δt_e) within a specific transmission window. Note that, the results are derived from time-truncated blocks of a single satellite overpass (13:22:17-13:27:18 UTC, September 19, 2016). The inset figure shows the proportion of SKL generated within the period starting Δtafter the start of the time window and ending Δt before the end of the time window. The gray area indicates the duration over which SKL generation is concentrated.

estimate the secret key length (SKL) corresponding to different effective transmission windows within that same pass. Figure 7 presents the finite-key SKL results for the Nanshan-Delingha OGS pair computed for arbitrary start-to-end time intervals (Δt_s to Δt_e) within this specific overpass, i.e., September 19, 2016, 13:22:17-13:27:18 UTC. To clearly illustrate the potential benefits brought by downlink scheduling optimisation, we adopt an idealized link within this section. The intrinsic loss and residual turbulence loss for two satellite-to-ground links is reduced from 9 dB to 2 dB. Note that this hypothetical assumption, and practical systems will require further technical iterations to approach this level of performance. All other sections of this work use the original 9 dB loss parameters.

The results show that no secret key was generated at the beginning or near the end of the transmission window. This is due to smaller data are received from the time with large average channel loss at lower elevations or longer ranges. Practically, secret key extraction can be optimized by truncating poorer quality data from the periods in these circumstances. This also suggests that the scarce time slots can be further released to design downlink schedule that allows the data collection for alternative possible OGS pairs.

With this in mind, we then design an optimal downlink schedule. Since the SatQKD keys are currently a very scarce resource, the objective of our downlink schedule optimisation is to maximize the total number of keys generated at the different OGS locations, considering the operational constraints of SatQKD tasks. Specifically, we consider the constraints: (1) no task is scheduled outside its predefined start and end times within its allowable transmission time window, ensuring a time-bound task execution duration; (2) at any instant, only one SatQKD task can utilize downlink resources, preventing temporal overlap; (3) a switching period is defined to account for the overhead required to reconfigure the payload (e.g., ATP system, quantum optics reset) when switching between different OGS pairs, enabling operational readiness between distinct tasks.

In optimisation formulation, we define a set of n candidate SatQKD tasks, and let s_i and e_i ($e_i > s_i$) denote the earliest start time and the end time for the task i, respectively. For time t (in seconds) in the planning horizon T, we use a binary variable $x_{i,t}$ to enforce the discrete decision on the scheduling task i. Given the SKL k_i generated upon successful execution of task i, we could therefore formulate the optimisation problem as an integer linear program (ILP) to maximize the SKL

$$\max \sum_{i=1}^{n} \sum_{t=s_i}^{e_i-d_i+1} \ell_{\mathrm{SKL}_i} \cdot x_{i,t} \tag{3}$$

where $d_i = e_i - s_i$ denotes the duration of task I, and the objective of the optimisation problem is to find a schedule that can maximize the total number of keys. In our modeling, the required switching period between different tasks is considered to be 10 seconds. Despite this optimistic parameterization, our work aims to explore the potential advantages of the implementations (see 'Method' section 'Optimisation problem' for the formulation and implementation details).

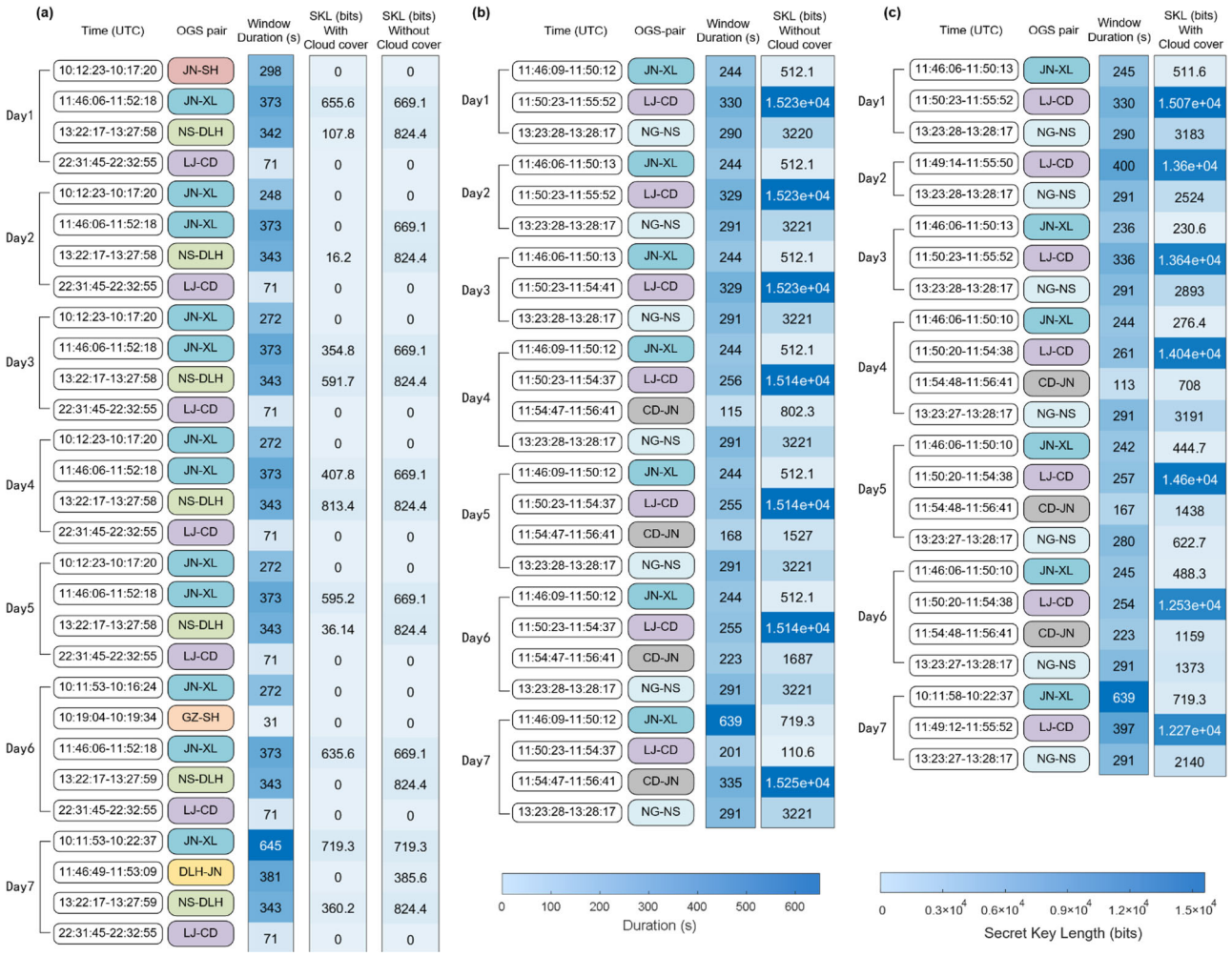


Fig. 8 | Illustration of the SSO Satellite SatQKD downlink schedule tasks during the week of Sep 22, 2016 UTC. Details include task start/end times (in UTC), target OGS pairs, task durations, and the expected SKL under four schedules. These schedules represent the results obtained via **a** non-optimized scheduling in the conditions of cloud cover/no cloud cover, and optimized scheduling in the conditions of **b** no cloud cover **c** cloud cover, respectively. JN, Jinan; SH, Shanghai; NS, Nanshan; DLH, Delingha; LJ, Lijiang; CD, Chengdu; XL, Xinlong; GZ, Guangzhou; NG, Ngari; HF, Hefei. The different colors used for OGS labels serve as visual distinction to identify different OGS sites.

Figure 8 summarizes the 7-day SSO SatQKD downlink schedules, with details of start/end times, durations, target OGS pairs, and expected SKLs. For a fair comparison, four schedules are included, corresponding to optimized and non-optimized scheduling under the conditions with and without cloud cover, respectively. For the non-optimized schedule, tasks are scheduled sequentially with non-overlapping transmission time windows to avoid interruptions. By comparing the SatQKD tasks of the Nanshan-Delingha (denoted NS-DLH) OGS pair from 13:22:17 to 13:27:58 Universal Time Coordinated (UTC), it can be seen that the SKL varies under different cloud cover disturbances, ultimately leading to a reduction of 70% in the total SKL. The schedule strictly prohibits any interruptions and re-scheduling during an active transmission time window, ensuring full execution of each window time before subsequent tasks are scheduled. However, there are cases of zero-key generation. This is due to task execution periods that suffer from poor quality data or entire communication windows characterized by high average link loss. By optimizing scheduling, durations can be flexibly arranged under the limited satellite overpass time. In particular, the total SKLs increases 22 times in the condition of cloud disturbance. Therefore, by integrating meteorological data for future cloud conditions, our approach enables optimization of downlink scheduling for the execution of various SatQKD tasks, as well as mitigating an adverse effect on optical transmissions from overcast days.

However, the periodicity of SSO satellite orbits imposes a deterministic concentration of SatQKD opportunities within cyclic time-window sequences, predominantly favoring specific OGS pairs such as Jinan-Xinlong (JN-XL) and Nanshan-Lijiang (NS-LJ), as evidenced by their dominance in secret key generation across all executed tasks. To this end, we investigate the performance of downlink schedule optimisation for polar orbit satellites. For comparison, we also introduced the “simple schedule” task, which uses non-optimized scheduling during task execution, while imposing an additional restriction on each execution duration of not less than 100 s to avoid excessive downlink switching times and increase execution difficulty. As illustrated in Fig. 9, the allowance of interruption operations within these windows introduces opportunities for key generation in alternative OGS pairs. For example, the number of pairs of OGS in the task increases from 7 to 14, whereas the enhancement in total SKL remains suboptimal. In contrast, the optimized schedule ultimately yields a total increase of at least 3 times in key generation for 10 different OGS pairs. Specifically, for the JN-XL OGS pair, although the SKL generated within its 218 s truncated window is less than in the entire window from 16:31:24 to 16:35:01 UTC on the 5th day, it frees up task execution periods with poor quality data collection, thereby enabling the Jinan-Hefei (JN-HF) OGS pair

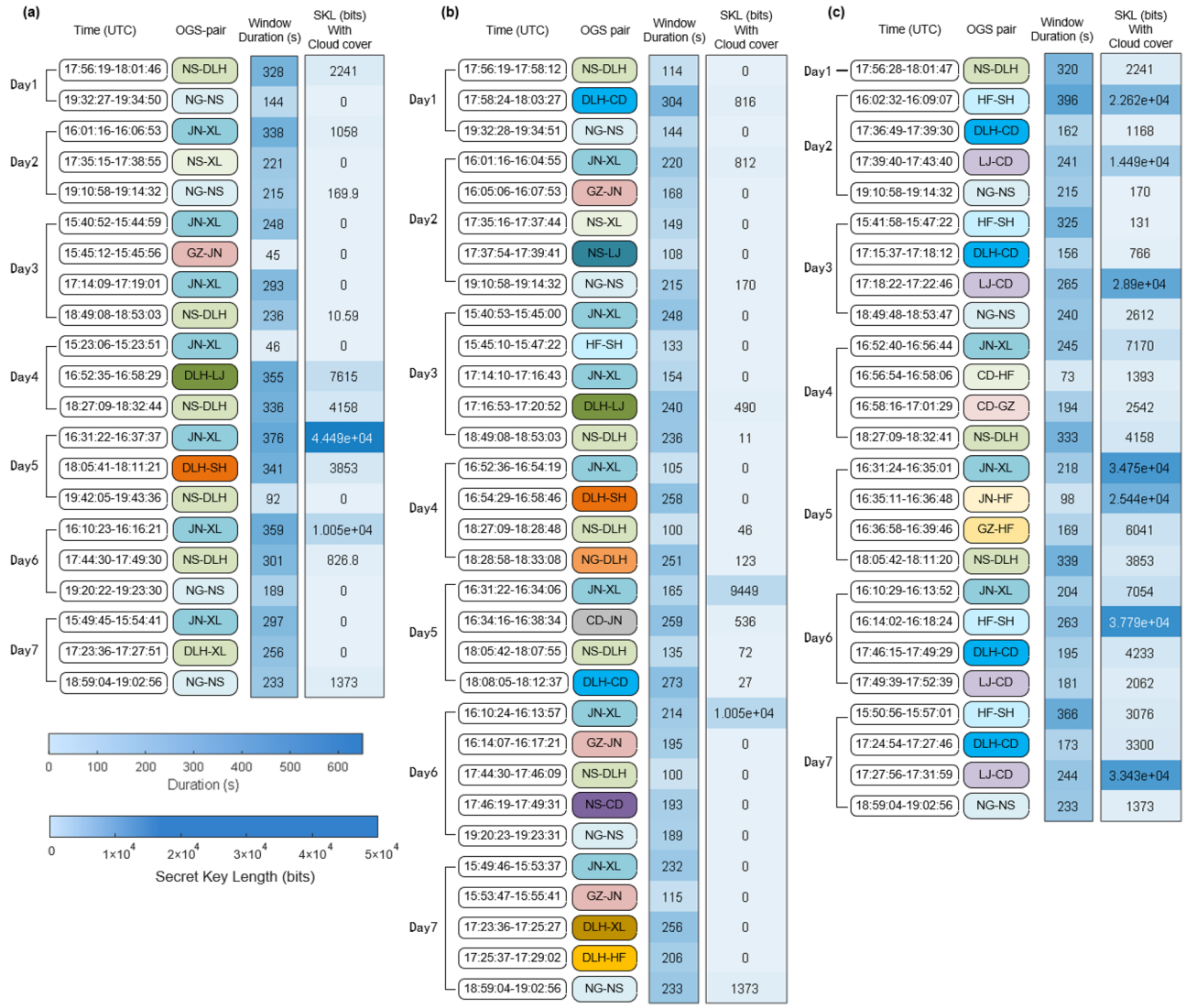


Fig. 9 | Illustration of the polar satellite SatQKD downlink schedule tasks during the week of Sep 22, 2016 UTC. Details include task start/end times (in UTC), target OGS pairs, task durations, and the expected SKL under three schedules. These schedules represent the results obtained via **a** non-optimized scheduling, **b** simple scheduling and **c** optimized scheduling, respectively. All schedules are carried out in the condition of cloud cover. JN, Jinan; SH, Shanghai; NS, Nanshan; DLH, Delingha; LJ, Lijiang; CD, Chengdu; XL, Xinlong; GZ, Guangzhou; NG, Ngari; HF, Hefei.

to be scheduled for SatQKD. Based on finite BBM92 SKL estimation with small block length, downlink schedule optimisation not only enables more OGS pairs but also ensures that each OGS pair conducts raw data collection during optimal periods within their allocated time windows. This indicates that key generation-oriented schedule optimisation promises a feasible way to improve efficiency in future global quantum communication networks.

Discussion

Satellite-based QKD (SatQKD) has emerged as a vital pathway for a global QKD network, with extensive experimental and engineering progress in single-downlink SatQKD advancing other SatQKD configurations. In practice, dual-downlink entanglement-based SatQKD faces distinct challenges that single-downlink modeling frameworks cannot address. We demonstrate that dual-downlink configurations involving one satellite and a pair of optical ground stations (OGSs) that feature exclusive satellite overpass geometries, potentially asymmetric ground system-optics parameters, and orbit-dependent metrics determine secret key length (SKL) generation. It is therefore of practical significance to study the dual-downlink SatQKD performance with a comprehensive modeling to include site-dependent effects, to reflect entanglement-based SatQKD performance dependence on ground system parameter symmetry, and to provide trade-offs for SKL extraction methods for different orbits. Further, the scenario where a quantum satellite is within the range of multiple OGS pairs introduces a downlink assignment challenge for future large constellations and remains underexplored, particularly with regard to finite data block statistics and changing cloud cover. We address this by optimizing the SatQKD downlink schedule with finite BBM92 key analysis, exploring the advantages in SKL generation efficiency and flexibility for global networking.

First, we advance a modeling framework integrating Sun-synchronous orbit (SSO) satellite orbital propagation, cloud cover quantification, and entanglement-based QKD channel analysis, to address the lack of orbital design metrics for dual-downlink SatQKD configurations, the limitation of clear-sky assumptions for application exploration, and the characterization of measurement data collection in dynamic channels. Incorporated with the Aerospace Tool Kit (ATK) software's orbit modeling, physical parameters, e.g., satellite-OGS ranges and elevations, are discretized at intervals to characterize time-averaged link loss changes. Critically, we propose a method to determine the ideal satellite overpass for a given OGS pair, enabling the longest transmission time window with lowest average two-link loss. Historical cloud cover data is integrated to quantify additional channel loss, providing a benchmark to support scheduling downlink SatQKD missions in those areas with clouds. Using entangled source and channel models, the instantaneous coincidence

detection rate and QBER are derived (Fig. 2f–h). We find that data collection conducted near the start or end of the communication window occurs at the cost of elevated QBER at low elevations, arising from poor parameter estimation. This suggests that gathering overpass data requires careful selection of time intervals and the identification of data segments detrimental to finite secret key generation. Further deep-learning predictions based on meteorological data or weather forecasts could be integrated to explore in advance the instantaneous SatQKD downlink channel fluctuations, and truncate adverse data segments by real-time adaptive thresholds.

Next, we characterize different overpass geometries via the descending node longitude (LDN) offset $\Delta\Omega$ to the ideal overpass traversing the midpoint of an OGS pair, which enables a generalized extrapolate performance to other dual-downlink SatQKD configurations. For the ground system specifications included in Fig. 3, we find that asymmetric extraneous count rates between the two downlinks induce a site-dependent impact on QBER that depends on the signal-to-noise ratio (SNR). In particular, as overpass geometry offsets move the satellite away from the OGS zenith, extraneous counts compound with higher link loss, reducing the SNR. Interestingly, the ideal overpass emerges as the optimal orbit configuration to maximize data block size, since data collection performance is independent of asymmetric link loss. This independence indicates the feasibility of the deployment of the SatQKD network with portable ground stations of varying apertures in remote regions. We clarify the distinct performance of two SKL extraction methods for different orbits. The *overpass-block SKL extraction* method, which aggregates data from identical overpasses on different days into a larger single processing block to achieve an asymptotic regime, is particularly applicable to low Earth orbit (LEO) satellites with daily repeatable overpasses, e.g., Sun and Earth synchronous orbit satellites. In practice, smaller and larger loss bins reveal a trade-off between statistical uncertainty and SKL performance, and excessive binning degrades secret key length, which suggests that the *Segmented-block extraction method* ultimately could evolve toward *Overpass-block extraction method*. Nonetheless, the *segmented-block SKL extraction* method offers a practical alternative when sufficient data is needed to generate a continuous secret key stream.

Finally, we highlight the significance of SatQKD downlink scheduling optimisation based on finite-size SKL estimation for a satellite performing SatQKD with multiple OGSs. We formulate the optimisation problem as an integer linear programming (ILP) problem in Eq. (3), which avoids cloud disturbance and supports the collection of raw data for other line-of-sight OGS pairs. For nominal one-week SatQKD missions with changing cloud cover, our optimized scheduling strategy avoids excessive link switching from time conflicts among numerous OGSs. By incorporating finite BBM92 key analysis into the optimisation, each OGS pair collects data during optimal intervals, with the adverse data collection periods flexibly truncated. This yields an increase in the expected total SKL of the SSO and polar satellites by 16 times and 10 times, respectively. While the results demonstrate that all missions considered here can improve the finite key generation, implementing on-demand scheduling dual-downlink SatQKD presents notable in-orbit engineering challenges, including but not limited to an efficient Acquisition, Tracking, and Pointing (ATP) mechanism, robust quantum network architectures, and dedicated protocols.

Our discussion of performance modeling, site-dependent effects, and scheduling of dual-downlink SatQKD opens the problem to further study. For a long-term vision of a satellite-based global quantum network, an extension would include a constellation of satellites that perform scheduling dual-downlink entanglement-based SatQKD to any two points, such as inter satellites, global ground stations, and high-altitude flying platforms[66,67]. As such, a comprehensive QKD channel model that considers daytime background light distributions, aircraft maneuverability, and atmospheric turbulent effects is necessary for deep-learning predictions enabling threshold selection and protocol parameters optimisation[68,69]. Further efforts should integrate orbit dependent metrics arising from different satellite constellations, such as communication durations, interval, and overpass geometries, to guide the finite key estimation. Additionally, exploiting scheduling with different constraint objectives could be developed for desirable SatQKD missions.

Methods

Integrated with satellite propagations to determine link geometry parameters

Here, we will discuss the way we define a transmission time window and how our modeling framework expects the link geometry parameters to be specified for the purposes of the instantaneous link loss calculation.

Different from single downlink SatQKD, dual-downlink SatQKD involves two line-of-sight OGSs. In addition to visibility, we impose current physical limitations, including (1) both OGSs have a minimum elevation angle for stabilization of the Acquisition, Tracking and Pointing (ATP) system, (2) the opportunity for SatQKD is restricted to periods when the satellite experiences a total eclipse in Earth's shadow (umbra) due to stray light restrictions. As such, we define the transmission time window as the specific period during which all the above conditions are satisfied, enabling dual downlink entanglement-based SatQKD. The time-varied link geometry parameters due to the satellite propagation can be predicted with SSO orbital dynamics. In our modeling, the transmission time window is discretized into time-slots, each with a duration of one second. Without loss of generality, the SSO satellite orbit is circular with an altitude of 500 km and the inclination of the altitude is 97.658°. Suppose that the latitude-longitude-altitude (LLA) coordinates of an OGS are known, using the SGP4 algorithm, the instantaneous ranges and elevation angles are computed at time steps of one second. Our work is integrated with Aerospace Tool Kit (ATK) software's orbit modeling capabilities[70], the calculation can be simplified, and the link geometry parameters change in intervals of one second to characterize time-varied downlinks with time discretization.

Using our modeling, we demonstrate the cumulative durations of transmission time windows aggregated in the week of Sep 22, 2016, Universal Time Coordinated (UTC) for the potential OGS pairs in Fig. 10. Intuitively, the transmission time for an OGS pair is shorter than that of any single OGS, and the adjacent OGSs always have the longest duration of the transmission time window. Owing to the physical limitations, the satellite was unable to perform entanglement-based SatQKD in some pairs of the OGSs. This result demonstrates the difference in the feasibility of performing SatQKD in these OGSs, which could be a valuable indicator to support decisions involving the locations of OGSs and the selection of the orbital parameters for a satellite.

Incorporated with meteorological data to simulate the dual-downlink channels

In our modeling, the link loss, including beam diffraction and atmosphere effects, can be regarded as a function of range and elevation angle, once the parameters related to the specifications of optics systems are fixed. A comprehensive analysis of link loss can also be found in ref. 61. Here, clear skies in the two localized areas are assumed in our modeling, therefore, the days with cloud disturbance are excluded from the execution of the SatQKD missions. As a result, much more variability is missed in the real-world results.

To approximate the cloud disturbance to transmission in terms of loss, we referred to meteorological information in Himawari-8 cloud statistics[71], in which the cloud optical thickness at each surface coordinate point was encoded as a discrete scale ranging from 0 to 150. Note that these data were grained on 10 min intervals, which is not precise enough for short periods of the transmission time window. To this end, we considered the cloud records at adjacent positions and randomly selected the cloud optical thickness value per second, with the range determined by their minimum and maximum values. Quantification is shown in Fig. 11. After linear mapping and normalization processing, a matrix within the range of 0–1 can be obtained. Then these results corresponding to the assumed OGS locations were assumed to be additional channel losses in the model. Kindly note that clouds introduce additional noise that might compromise the data collection. Modeling the dynamic background photon rate requires the decay of solar irradiance, the sun position relative to the ground, diffuse sky radiation, and usually the design specifications specified by the manufacturers. Work along these lines will be our future work.

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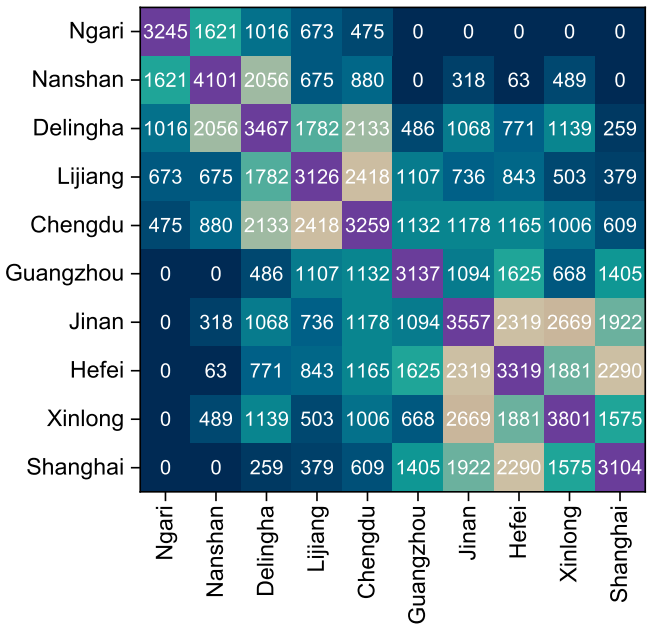


Fig. 10 | Duration of the transmission time windows observed for ten assumed OGSs locations during the week of Sep 22, 2016 UTC. For the axis of the matrix, the ten OGSs are arranged in order of longitude, from the west to the east.

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Ideal overpass geometry

The satellite trajectory passing through the great-circle midpoint of two OGSs promises the ideal overpass geometry, which provides the longest

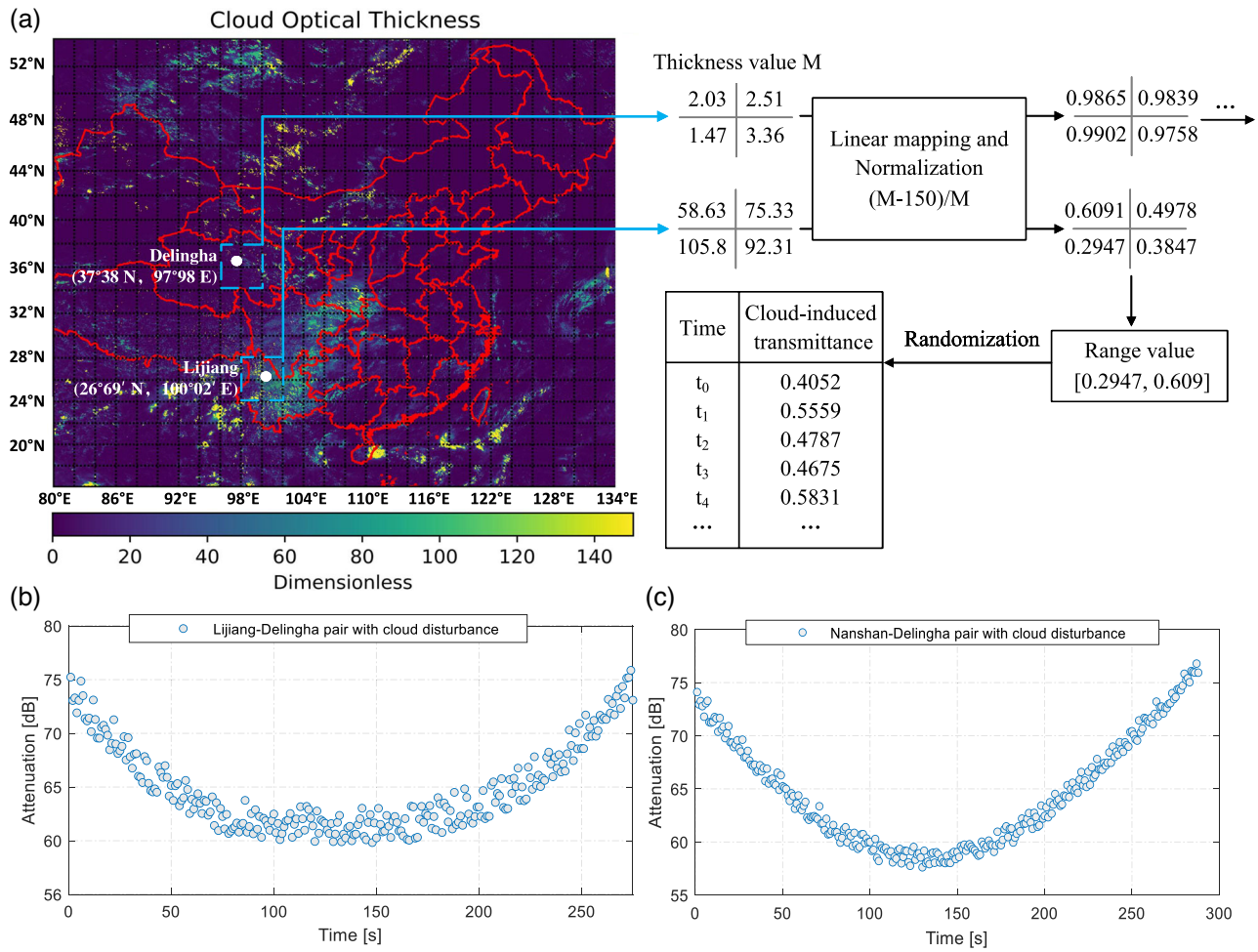


Fig. 11 | Implementations of a downlink channel simulation. a Cloud optical thickness over the assumed Chinese OGSs on 23rd Sep 2016, 16:32:40 UTC, which is then converted to additional channel losses. **b**, **c** With the cloud disturbance, the overall two-link loss fluctuations are simulated for Lijiang–Delingha and Nanshan–Delingha OGS pairs, respectively.

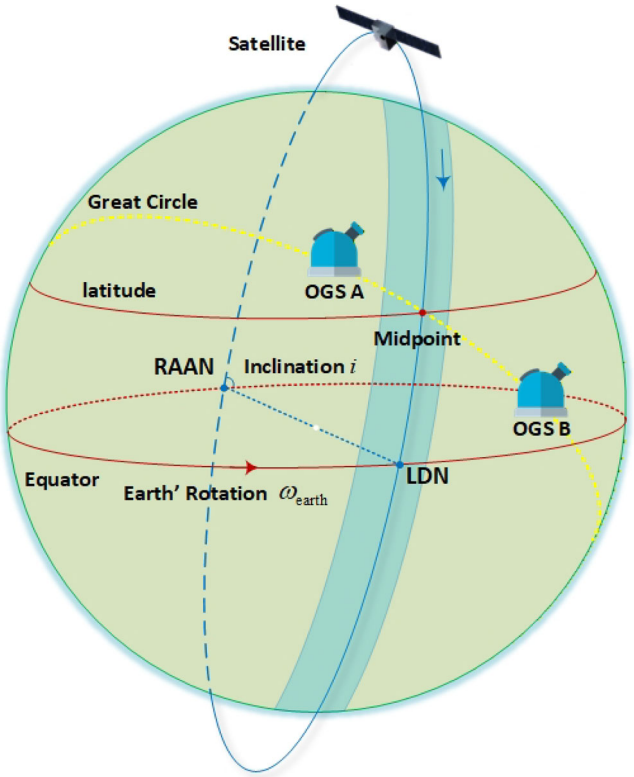


Fig. 12 | LDN-based model for ideal satellite-dual OGS overpass geometry. We consider a satellite in 500 km-altitude circular SSO with the inclination $i = 97.4065^{\circ}$. Earth is centered with its equatorial plane depicted by a red line, and the SSO orbital plane, colored blue, forms the inclination, indicating a retrograde orbit. The point RAAN marks the intersection where the orbit crosses from the southern to the northern hemisphere, and its longitude defines the RAAN, and thus the LDN 180° apart from the RAAN after accounting for Earth's rotation. With a target coordinate such as OGS A, the corresponding satellite trajectory that promises an ideal overpass geometry can be obtained by determining the LDN of the satellite. Here, a circular orbit (neglecting eccentricity) and a two-body system (ignoring perturbations) are assumed. The fundamental parameters include Earth's angular velocity ω_{earth} is 7.29×10^{-5}rad/s and gravitational constant ($\mu_0 = 3.9861 \times 10^5$ km^3/s^2), from which the orbital angular velocity ($v_g = 0.063^{\circ}$/s) is derived for a circular orbit of altitude 500 km. Incorporating elliptical orbit corrections, orbital perturbation compensations, and precise Earth rotation would be an interesting direction for orbit parameters design.

transmission time and the shortest average total range during the period due to spherical geometric symmetry. The yellow dashed line representing the great circle on a sphere defines the shortest ground path between two OGSs (Fig. 12). The SSO satellite maintains balanced proximity to both OGSs, thereby maximizing the overlap of their co-visible time and minimizing the relative distance to each OGS throughout the communication window.

In our work, the satellite trajectory that promises the ideal overpass geometry could be obtained by determining the LDN of the satellite. To this end, we present a general methodology for determining the LDN of an SSO satellite that traverses an OGS zenith (i.e., the specific latitude and longitude position), which combines the orbital geometry with Earth's rotation dynamics. Specifically, the true anomaly f at the target latitude is calculated via spherical trigonometry within the orbital plane

$$\sin(f) = \sin\left(\theta_{\text{Target-lat}}\right) / \sin(i) \tag{4}$$

where i is the angle of inclination of an SSO satellite. Thus, the time Δt_1 for an SSO satellite to travel from the Right Ascension of Ascending Node (RAAN) to the target is then derived by dividing the angular distance by the orbital velocity v_g, we can obtain $\Delta t_1 = (180^{\circ} - f)/v_g$. During time Δt_1, as indicated by the red arrow representing Earth's rotation, the target longitude shifts by

$$\Delta\theta_{\text{earth}} = \omega_{\text{earth}} \Delta t_1 \frac{180^{\circ}}{\pi} \tag{5}$$

where ω_{earth} represents Earth's angular velocity. As such, the RAAN is computed by adjusting the target initial longitude with this rotation-induced shift and geometric corrections from the orbital inclination, following the formula

$$\begin{aligned}\Omega_R = \theta_{\text{Target-lat}} - \arctan\left[\cos(i)\tan(180^{\circ} - f)\right] \\ 0.5em + \Delta\theta_{\text{earth}} + 180^{\circ}\end{aligned} \tag{6}$$

The LDN is subsequently derived by considering the satellite's half-orbit transit (i.e., taking time $\Delta t_2 = 180^{\circ}/v_g$ from the RAAN. Given the additional Earth rotation during Δt_2 subtracted from Ω_R, we have that

$$\Omega_D = \Omega_R - \omega_{\text{earth}} \Delta t_2 \frac{180^{\circ}}{\pi} - 180^{\circ} \tag{7}$$

Note that our approach assumes a circular orbit and neglects minor perturbations, which is applicable to SSO satellites with near-circular trajectories and provides a methodology for ideal overpass geometry selection and ground station coverage analysis.

Finite key analysis of entanglement-based BBM92 SatQKD

In the following, we consider photon transmission in entanglement-based SatQKD. The entangled photon pair is generated in the onboard optics via pump pulse spontaneous parametric down conversion (SPDC) source. Then, the polarization-entangled photons are split and transmitted to two parties, Alice (A) and Bob (B). As noted above, a satellite overpass experiences highly varying channel loss, resulting in a remarkable change in the detection rates and the QBER. Combined with our dual downlink SatQKD model, we consider instantaneous link loss η_a and η_b to be the overall detection efficiencies for Alice and Bob, respectively. Here, η_a and η_b include all the losses owing to the satellite-to-ground quantum communication channel, coupling loss, and detection inefficiency. Using the standard source model, if the probability of generating n photon pairs is P_n and the coincidence detection probability of n photon pairs is Y_n, we can calculate the overall coincidence probability per pump pulse when employing non-photon-number-resolving detectors

$$\begin{aligned}Q = \sum_{n=0}^{\infty} P_n Y_n = 1 - \frac{1 - p_{ec_a}}{(1+\eta_a\gamma)^2} - \frac{1 - p_{ec_b}}{(1+\eta_b\gamma)^2} \\ + \frac{(1 - p_{ec_a})(1 - p_{ec_b})}{(1+\eta_a\gamma+\eta_b\gamma-\eta_a\eta_b\gamma)^2}\end{aligned} \tag{8}$$

where γ is related to the pump power of the laser $2\gamma = \mu$ (which denotes the mean number of pairs per pump pulse) and $p_{ec_a}(p_{ec_b})$ is the noise or background count observed in OGS A (or OGS B) per pump pulse. In this work, we set $\mu = 0.053$.

After transmission, we assume post-selection on coincident events, along with satellite-to-ground quantum communication channel assumptions. The post-selection allows one to discard any occurrence where one site registers a photon and the other does not. Considering the high channel loss and background noise in entanglement-based SatQKD, we include the effects of measurement system errors, background light, and photon loss (i.e., potential multiple photon pair occurrences). To estimate the probability of detecting an error per pump pulse, we refer to ref. 63 where the expression for the estimated total error probability per pump pulse is

$$Q_{\text{est}} \leq \frac{E_b + E_{|\Phi^n\rangle} + E_{\text{MPE}}}{R} \tag{9}$$

Here, E_b is the background-induced error, for which detects a background count and Bob (Alice) detects at least one source photon, or both loses entangled photons. Then, E_{MPE} is the measurement-imperfect errors in the detection and polarization-analysis systems, including the ideal case

where one photon pair is detected at one side. This stems from imperfections in the measurement system in which multiple source photons are detected simultaneously on different detectors of the one OGS side. $E_{|\Phi^n\rangle}$ is the multi-pair induced errors. Errors that occur when two or more photon pairs are produced within one pump pulse and detected within the detector's coincidence window. This raises a case where at least two photon pairs are created, and at least one photon on each side is detected (possibly not from the same pair, which directly influences the count rate).

We remark that for a CW-pumped source, high-visibility two-photon Hong-Ou-Mandel (HOM) interference is achievable without strict spectral filtering. By contrast, pulsed-pumped photon-pair sources typically yield high photon purity only near the group-velocity-matched (GVM) wavelength[72,73]. Under equal total pump power, pulsed pumping delivers far higher instantaneous peak power, leading to considerably larger multi-pair generation probability compared with CW-pumped counterparts. Featuring excellent stability and sustained source brightness, CW-pumped sources could be also suitable for compact satellite platforms. We consider the two schemes complementary, and will advance entanglement-based SatQKD practical engineering implementation.

Having the quantum channel and source models, we then denote the lengths of the sifted events by $L = f_s Q$ with the total error rate Q_{est}. Then, the size of the raw key randomly sampled from the sifted events is fixed to some positive integer $L = l + k$, where $k = L(1 - r)$ is the amount of raw key used for parameter estimation, $l = Lr$ is the amount of raw key left for key generation, and r is the security parameter estimation ratio. Following standard security definitions, we assume that the QKD protocol is ε secure if it is both ε_{sec} secret and ε_{cor} correct. Here, we set $\varepsilon_{\text{sec}} = 10^{-9}$, and $\varepsilon_{\text{cor}} = 10^{-12}$. Assuming that the measurements on two OGS receivers are mutually unbiased (e.g., no polarization misalignment), we can use the quantum leftover hash lemma to derive a bound[74] on the extractable SKL (denoted by ℓ_{SKL}) for a given post-processing block size

$$\ell_{\text{SKL}} = \max_{\zeta \in (0, \varepsilon_{\text{sec}}/4]} \lfloor l_{\text{ext}} + 4\log_2 \zeta - 2 \rfloor \tag{10}$$

and

$$\begin{aligned} l_{\text{ext}} =& l\left(1 - h_2\left(Q_{\text{est}} + \Delta(l, k, \zeta)\right)\right) \\ & - \lambda_{\text{EC}} - \log_2 \frac{2}{\varepsilon_{\text{cor}}} \end{aligned} \tag{11}$$

where

$$\lambda_{\text{EC}} = 1.12 l h_2(Q_{\text{est}}), \tag{12}$$

$$\Delta(l, k, \zeta) \equiv \sqrt{\frac{l + kk + 1}{lk \quad k} \ln \frac{1}{\zeta}}, \tag{13}$$

$$h_2(x) \equiv -x\log_2(x) - (1 - x)\log_2(1 - x) \tag{14}$$

is the binary entropy, Δ is the statistical noise due to finite statistics of the amount of raw key used for parameter estimation k and the raw key left for key generation l, and λ_{EC} and $\log_2 \frac{2}{\varepsilon_{\text{cor}}}$ are information leakages due to error correction and verification, respectively. Here, ζ is a constrained parameter to be optimized, which would be used to determine the smooth min-entropy of the raw key X (of size L) given E (Eve's total information about the QKD system). Thus, l_{ext} is the maximum theoretical length that one can extract, while $\ell_{\text{SKL}} \leq l_{\text{ext}}$ is the final output of the QKD system in practice (For more details, see ref. 75). Combined with the two SKL extraction methods in the main text, the Pseudocode for the finite SKL is illustrated in **Algorithm 1**.

Algorithm 1. **Pseudocode of SKL extraction methods**

$f_s \leftarrow$ number of polarization-entangled photon pairs transmission
$N \leftarrow$ number of satellite overpasses during one year
$[\beta_A(t), \beta_B(t)] \leftarrow$ elevationfromthesatellitetoOGSA(OGSB)attime t
$[R_A(t), R_B(t)] \leftarrow$ rangefromthesatellitetoOGSA(OGSB)attime t
for all $t = \Delta t_{\text{start}}$ to Δt_{end} **do**
  $\eta_a, \eta_b \leftarrow f[\beta_A(t), R_A(t)], f[\beta_B(t), R_B(t)]$
  $Q[\eta_a, \eta_b] \leftarrow$ overallcoincidenceprobabilityperpumppulseattime t
  $Q_{\text{est}}[\eta_a, \eta_b] \leftarrow$ errorcountprobabilityperpumppulseattime t
end for
———*Method 1: Overpass-block SKL extraction*———
$n \leftarrow$ fixed satellite overpasses per day
$T \leftarrow$ number of aggregation days
for all $i = 1$ to n **do**
  $s_i^{\text{overpass_data}} \leftarrow \sum_{t=\Delta t_{\text{start}}}^{\Delta t_{\text{end}}} f_s Q[\eta_a, \eta_b]$
  $e_i^{\text{overpass_error}} \leftarrow \sum_{t=\Delta t_{\text{start}}}^{\Delta t_{\text{end}}} f_s Q_{\text{est}}[\eta_a, \eta_b]$
  $\text{QBER}_i \leftarrow \frac{e_i^{\text{overpass_error}}}{s_i^{\text{overpass_data}}}$
end for
$\sum_{i=1}^{n} \ell_{\text{SKL}}[T(s_i^{\text{overpass_data}}, e_i^{\text{overpass_error}})] \leftarrow$ SKLextractedfromtheblocksoverTdays
$\ell_{\text{year}} = \frac{365}{T} \sum_{i=1}^{n} \ell_{\text{SKL}}[T(s_i^{\text{overpass_data}}, e_i^{\text{overpass_error}})] \leftarrow$ expectedannualSKL
———*Method 2: Segmented-block SKL extraction*———
$P \leftarrow$ number of aggregation overpasses
$m \leftarrow$ number of data blocks segmented based on the 1-dB loss-bins of the overpasses$P$
for all $j = 1$ to m **do**
  $s_j^{\text{segmented_data}} \leftarrow f_s Q[\eta_a, \eta_b],\ \eta_j < \eta_a + \eta_b \leq \eta_{j+1}$
  $e_j^{\text{segmented_error}} \leftarrow f_s Q_{\text{est}}[\eta_a, \eta_b],\ \eta_j < \eta_a + \eta_b \leq \eta_{j+1}$
  $\text{QBER}_j \leftarrow \frac{e_j^{\text{segmented_error}}}{s_j^{\text{segmented_data}}}$
end for
$\ell_{\text{SKL}}(s_j^{\text{segmented_data}}, e_j^{\text{segmented_error}}) \leftarrow$ SKL extracted from the blocks over P overpasses
$\ell_{\text{year}} = \frac{N}{P} \sum_{j=1}^{m} \ell_{\text{SKL}}\left(s_j^{\text{segmented_data}}, e_j^{\text{segmented_error}}\right) \leftarrow$ expected annual SKL

Optimisation problem

Our optimisation problem is formulated as an ILP with a linear objective function, linear constraints, and binary variables. The goal is to determine the optimal downlink schedule such that transmission durations are allocated to transfer raw data to a given OGS pair. As such, the total number of keys generated at the different OGS locations can be maximized, under the operational constraints of the downlink SatQKD to large OGS numbers. The optimisation model is formulated below.

$$\max \quad \sum_{i=1}^{n} \sum_{t=s_i}^{e_i-d_i+1} \ell_{\mathrm{SKL}_i} \cdot x_{i,t} \qquad \text{s.t. } x_{i,t} \in \{0,1\} \quad \forall i,t \tag{15}$$

$$\sum_{t=s_i}^{e_i-d_i+1} x_{i,t} \le 1 \quad \forall i \in n \tag{16}$$

$$\sum_{i=1}^{n} \sum_{\substack{t \le t' \\ t'<t+d_i}} x_{i,t} \le 1 \quad \forall t' \in T, \forall i \in n \tag{17}$$

$$t' \ge t + d_i + c(1 - \delta_{g_i g_j}) \quad \forall i,j \in n,\ i \ne j \tag{18}$$

In the set of n candidate tasks, s_i and e_i ($e_i > s_i$) denote the earliest start time and the end time for task i, respectively. In the duration $d_i = e_i - s_i$ of task i, we use k_i to represent the SKL yield for task i. For time t (in seconds) in the planning horizon T, we use a binary variable $x_{i,t}$ to enforce the discrete decision on the scheduling task i. In constraint (15), the decision variables are binary to enforce the scheduling of discrete tasks, where $x_{i,t} = 1$ if task I starts at time t, and 0 otherwise. The constraint (16) bounds the task feasibility, for which no task is scheduled outside its predefined start and end times or executed more than once. The constraint (17) ensures that, at any time t, only one task can utilize the SatQKD downlink, preventing multiple tasks simultaneously executed. Finally, the constraint (18) declares that if two consecutive tasks j and i belong to different OGS pair tasks, i.e., $g_i \neq g_j$, a switching period c must elapse between the end time of task i and the start time of j.

Data acquisition and pre-processing involve ATK orbital simulations and quantum channel modeling to capture SatQKD performance. Four datasets are derived: a binary visibility matrix $\mathbf{X}$ of dimension $45 \times 604{,}800$ across 45 OGS pairs over 604,800 seconds, a raw data matrix D incorporating detection rates and QBER, a cell array $\mathbf{K}$ storing secure key yields per time window. These datasets are structured into inputs for the schedule optimization, ensuring representation of physical constraints and key generation capabilities. The model is resolved using a dynamic programming framework, for which we consider binary search to retrieve the predefined tasks, utilize state transitions to select the optimal output path, and reconstruct the optimal schedule through backtracking. The program exhibits a time complexity of a near linear $O(n \log n)$. For more details on the implementation, see the source code developed for this study.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to ongoing analyses and the inclusion of unpublished follow-up work, but are available from the corresponding author on reasonable request.

Code availability

The results are reproducible with the code available on GitCode https://gitcode.com/qq_24405641/Code_for_SatQKD_simulation.

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Author contributions

Xingyu Wang and Jiahao Li conceived the core research idea, developed the dual-downlink SatQKD comodeling framework, and performed the simulations, with additional assistance from Bolong Wang. Yuexiang Cao and Qian Li conducted comprehensive literature reviews and assisted with data validation. Yonghui Li and Haoran Hu supported the cloud cover quantification modeling. Xingyu Wang and Lei Shi obtained funding for initiated the research. Chen Dong, Nan Xiao and Kai Guo supervised the research direction and participated in discussions on the results. Xingyu Wang and Jiahao Li contributed equally to this work.

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Competing interests

The authors declare no competing interests.

Additional information

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